

6.3.5 Activity

Context A – Vacation Savings

Clarita wants to save at least \$600 to go on a vacation this summer. Clarita knows that she will not need more than \$825. She currently has \$97.50 in her bank account and she makes \$15 per hour by tutoring. She can work any fraction of an hour, if needed. Clarita wants to determine the amount of time she could work to allow her to save enough for her vacation.

Context B – Non-Operating Temperatures

In the specifications for its products, Apple distinguishes between the operating temperatures, or the range of temperatures where the product will work, and the non-operating temperatures, where the product may not work, but will not break. The non-operating temperature for the iPhone 11 in degrees Celsius, °C, is -25°C to 45°C . Determine the range for non-operating temperature for the iPhone 11 in degrees Fahrenheit, °F. The formula $C = \frac{5}{9}(F - 32)$ can be used to convert from Celsius to Fahrenheit.

Context C – Gym Memberships

Malika is looking for a gym to join to take kickboxing classes. She is considering two options.

Athletic Club of Tampa	Bayshore Gym
• Monthly fee of \$15 • Additional \$8 per class	• Monthly fee of \$22 • Additional \$6.75 per class

Malika wants to know for how many classes the Athletic Club of Tampa is cheaper than Bayshore Gym.

Context D – Planting Citrus Trees

Hai is preparing a square area to plant orange and lemon trees. He needs to purchase fencing to go around the area, as well as an additional 15.2 yards (yd) of fencing to separate the two types of trees. In total, he wants to purchase at least 60 yd and no more than 100 yd of fencing. Hai needs to determine the possible side lengths, s , of the square area that will allow him to stay within that range of fencing.